

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

10 Logic and Proof (lp15)

Propositional
Logic, sequent
calculus

- (a) A mysterious propositional connective, $\odot$, has the following sequent calculus rule, $(\odot l)$:

$$\frac{A, B, \Gamma \Rightarrow \Delta \quad \Gamma \Rightarrow \Delta, A, B}{A \odot B, \Gamma \Rightarrow \Delta}$$

Present the corresponding right-side sequent calculus rule, $(\odot r)$, along with the truth table for $\odot$. [5 marks]

Answer:

$$\frac{A, \Gamma \Rightarrow \Delta, B \quad B, \Gamma \Rightarrow \Delta, A}{\Gamma \Rightarrow \Delta, A \odot B}$$

So $\odot$ is logical if-and-only-if ($\leftrightarrow$), with the corresponding truth table.

First-order Logic,
sequent calculus

- (b) For the following formula, either exhibit a formal proof (using the sequent calculus, augmented with the $(\odot l)$ rule above) or exhibit a falsifying interpretation:

$$\forall x(P(x) \odot Q(x)), \exists x P(x) \Rightarrow \exists x Q(x)$$

[5 marks]

Answer: Here is a sequent calculus proof. The inference at the bottom must be $(\exists l)$.

$$\frac{\frac{\frac{\frac{\frac{P(x), Q(x), P(x) \Rightarrow Q(x)}{P(x) \odot Q(x), P(x) \Rightarrow Q(x)} (\odot l)}{P(x) \odot Q(x), P(x) \Rightarrow \exists x Q(x)} (\exists r)}{\forall x(P(x) \odot Q(x)), P(x) \Rightarrow \exists x Q(x)} (\forall l)}{\forall x(P(x) \odot Q(x)), \exists x P(x) \Rightarrow \exists x Q(x)} (\exists l)$$

propositional
logic, DPLL
procedure

- (c) Use the DPLL method to find **all** models (if any) satisfying the following set of formulas.

$$\begin{aligned} (P \wedge R) &\rightarrow Q \\ (\neg Q \wedge R) &\rightarrow P \\ Q &\rightarrow (P \vee R) \\ \neg(P \wedge Q \wedge R) \\ P &\rightarrow R \end{aligned}$$

[5 marks]

Answer: First, convert to clauses:

$$\{\neg P, \neg R, Q\} \quad \{P, Q, \neg R\} \quad \{P, \neg Q, R\} \quad \{\neg P, \neg Q, \neg R\} \quad \{\neg P, R\}$$

Now if P is true then we get

$$\{\neg R, Q\} \quad \{\neg Q, \neg R\} \quad \{R\}$$

And then (by unit propagation with R)

$$\{Q\} \quad \{\neg Q\}$$

and then the empty clause. Therefore, this case is inconsistent.

But if P is false then we get

$$\{Q, \neg R\} \quad \{\neg Q, R\}.$$

A further case split is required, when if Q is true then so is R ; moreover, if Q is false then so is R .

DPLL shows us that P must be false. Then Q and R must have the same truth value, so there are precisely two models: $P = Q = R = \text{false}$ and $P = \text{false}, Q = R = \text{true}$.

modal logic

- (d) The modal logic S5 differs from S4 in requiring the accessibility relation to be symmetric, as well as reflexive and transitive. Present a formula that is valid for S5 but not for S4. Explain why it is valid for S5 and demonstrate that it is not valid for S4 by drawing an S4 modal frame for which it fails. [5 marks]

Answer: The simplest answer is $A \rightarrow \Box \Diamond A$, which is Axiom B in the notes. It is valid for S5 because if some formula A is true in some world w , then in any world v accessible from w we can return to w — because accessibility is symmetric — and make A true. It is falsified in any S4 modal frame where A is true in some world w and false in some accessible world v where there is no route back to w .

